

# Participation Report: ATRIUM "Computer Vision in Archaeology" Training School

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## Introduction

Hosted by the Institute of Archaeology of the Czech Academy of Sciences, the ATRIUM "Computer Vision in Archaeology" training school held in Brno (14–18 September) was honestly one of the most useful (and fun) experiences of my PhD so far.

As a PhD candidate in heritage science, my research focuses on non-invasive diagnostics, involving extensive work with spectroscopic and imaging data. Attending this school was also a fantastic opportunity to engage more actively with the broader MAIA network (COST Action CA23141). Within MAIA, I belong to Working Group 2 (WG2), which focuses on "Digital Comparative Collections and AI Training Data for Archaeology". Because WG2 addresses key questions around how heritage datasets are documented, accessed, and made "AI-ready," the training school felt like a very natural extension of my WG2 involvement, effectively bridging raw material diagnostics with computational workflows. The week absolutely delivered: it gave me a solid refresher on the basics, threw me out of my comfort zone with new tools, and offered plenty of fresh ideas to bring back to the lab.

## Program

While I already had some basic Python experience from my research, going back over the coding fundamentals was really helpful. It refreshed my knowledge, cleared up a few gaps, and gave me a much stronger footing before we jumped into the technical topics of the week. On the other hand, learning to "code with agents" was quite new territory for me. Discovering how to use AI assistants to help write, debug, and optimize code was a bit of a game-changer.

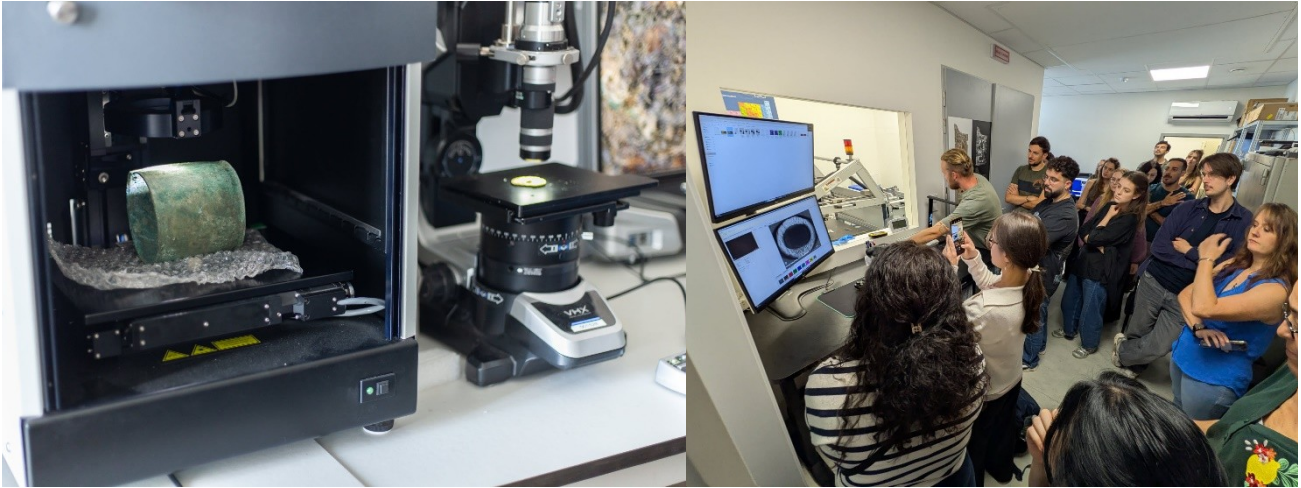
Instead of just talking about theory, we got our hands dirty with real archaeological datasets:

- ◇ Working on image standardisation, OpenCV masks, and data augmentation for microscopic use-wear traces felt very relevant to my background. Pre-processing image data before analysis is a big part of my own work, but learning specific OpenCV tools for masking gave me some practical new tricks for handling my own images.
- ◇ Jumping from microscopic details to satellite imagery and remote sensing was very interesting: seeing how computer vision scales up to landscape archaeology really broadened my perspective.
- ◇ The final sessions on large language models (LLMs), vision-language models, and attention maps were deeply relevant. We had a great discussion on bias, reproducibility, and the limits of "black-box" models. Since my PhD relies on machine learning to interpret complex data, this gave me a much clearer critical framework to ensure my own models stay transparent and scientifically sound.



## Into the lab (and the city)

One of my favorite parts of the week didn't happen in front of a computer screen. We got to tour the Institute's research laboratories and archives. Seeing how physical artifacts are curated and analyzed before they ever become a digital image was a fantastic reality check. It nicely grounded all the coding we were doing back in the material reality of cultural heritage.



Of course, the social side made the week complete. The organizers struck a great balance between work and relaxing. The Monday ice-breaker, the walking tour through the historic center of Brno, and the Wednesday evening social event were brilliant. It turns out that complaining about buggy code or dataset licensing over a drink is the single best way to network! I left with great contacts and a real sense of community.



## Final thoughts

Much of why the week worked so well was down to the incredible team of instructors. They were genuinely fantastic throughout, each brought their own unique background and expertise to the table, while remaining incredibly approachable and supportive whenever any of us got stuck in the code.

Looking back, I am genuinely glad I took part in these five days. The school managed to be technically rich without ever losing sight of the actual heritage questions that matter. I would recommend this training school without hesitation to anyone working at the crossroads of cultural heritage and computational methods. Plus, Brno is a fantastic city to explore!

